INCH-POUND

The documentation and process conversion measures necessary to comply with this document shall be completed by 21 April 2021.

MIL-PRF-19500/297K
W/AMENDMENT 1
21 December 2020
SUPERSEDING
MIL-PRF-19500/297K
24 January 2020

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
STANDARD AND REVERSE POLARITY,
TYPES 1N1184, 1N1186, 1N1188, 1N1190, 1N3766, 1N3768,
QUALITY LEVELS JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for silicon semiconductor power rectifier diodes. Three levels of product assurance (JAN, JANTX, and JANTXV) are provided for each device as specified in MIL-PRF-19500.

1.2 Package outline. The device package outline is a DO-203AB (formerly DO-5) in accordance with figure 1.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25 °C.

Type	V_R	V_{RM}	I_O (1) T_C = 150 °C	I_f (surge) at 1/120 s T_C = 150 °C	Barometric pressure (reduced)	$R_{\theta JC}$	Storage temp. T_C	Junction temp. T_J
	V dc	V (pk)	A dc	A	mmHg	°C/W max	°C	°C
1N1184, R	100	100	35	500	8	0.8	−65	−65
1N1186, R	200	200	35	500	8	0.8		
1N1188, R	400	400	35	500	8	0.8	to	to
1N1190, R	600	600	35	500	16	0.8		
1N3766, R	800	800	35	500	30	0.8	+175	+175
1N3768, R	1,000	1,000	35	500	54	0.8		

(1) Derate linearly 1.4 A dc/°C between T_C = +150 °C to T_C = +175 °C.

1.4 Primary electrical characteristics. Primary electrical characteristics are not applicable for this specification sheet.

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1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level. Quality level designator "JAN" is the only level that is applicable for this specification sheet.

1.5.2 Device type. The designation system for the devices covered by this specification sheet is as follows.

1.5.2.1 First number and first letter symbols. The devices of this specification sheet use the first number and letter symbols "1N".

1.5.2.2 Second number symbols. The second number symbols for the devices covered by this specification sheet are as follows: "1184", "1186", "1188", "1190", "3766" and "3768".

1.5.3 Suffix symbols. The following suffix symbol(s) are incorporated into the PINs for this specification sheet.

	A blank suffix indicates a standard polarity package of the diode as described by 3.4.3.
R	An R suffix indicates reverse polarity packaging of the diode as described by 3.4.3.

1.5.4 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

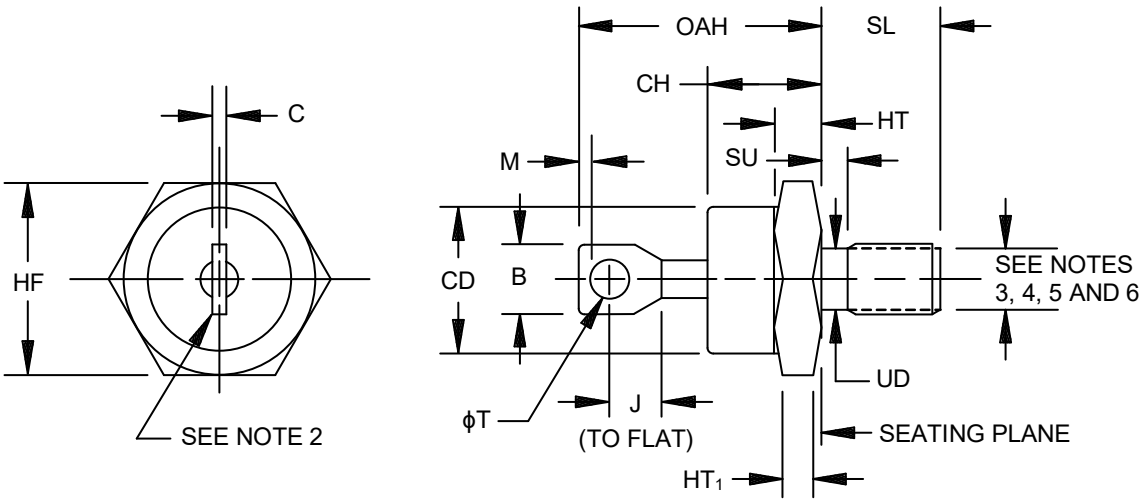
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions				Symbol	Dimensions			
	Inches		Millimeters			Inches		Millimeters	
	Min	Max	Min	Max		Min	Max	Min	Max
B	.250	.375	6.35	9.52	J	.156		3.97	
C		.080		2.03	M	.030		0.77	
CD		.667		16.94	OAH	.750	1.000	19.05	25.40
CH		.450		11.43	SL	.422	.453	10.72	11.50
HF	.667	.687	16.95	17.44	SU		.090		2.29
HT	.115	.200	2.93	5.08	UD	.220	5.59	.249	6.32
HT_1	.060		1.53		ϕT	.140	.175	3.56	4.45

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Angular orientation of this terminal is undefined. Terminal shape is unrestricted. See 3.4.3 for polarity.
3. The ASME thread reference is .250−28 UNF−2A.
4. The maximum pitch diameter of plated threads shall be basic pitch diameter .2268 (5.76 mm).
5. Complete threads to extend to within 2.5 threads of seating plane.
6. Units must not be damaged by torque of 30 inch-pounds (3.39 N m) applied to .250−28 UNF−2B nut assembled on thread.
7. A chamfer or undercut on one or both ends of the hex portion is optional. The minimum base diameter at seating plane is .600 inch (15.24 mm).
8. In accordance with ASME Y14.5M, diameters are equivalent to Øx symbology.

FIGURE 1. Physical dimensions for stud mounted case DO-203AB package (all device types)(formerly DO-5).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. The Interface and physical dimensions shall be as specified in MIL-PRF-19500 and figure 1 (DO-203AB, formerly DO-5).

3.4.1 Lead finish. Unless otherwise specified, lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Diode construction. All devices shall be in accordance with the requirements of MIL-PRF-19500.

3.4.3 Polarity. For standard polarity devices, the anode is connected to the terminal and the cathode is connected to the case (stud). For reverse polarity devices, the cathode is connected to the terminal and the anode is connected to the case (stud).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500. The polarity shall be indicated by a graphic symbol with the arrow pointing toward the negative end for forward bias.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3 and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in table I herein.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I, II, and III).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein. Tests in either polarity shall be sufficient to obtain qualification approval of both polarities.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (quality levels JANTXV and JANTX only). Screening shall be in accordance with appendix E, table E-IV of MIL-PRF-19500 and as specified herein. Specified electrical measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurements for quality levels JANTXV and JANTX
1a	Not required
1b	Required (JANTXV only)
2	Not required
3a	Required
3b	Surge current (see 4.3.1)
(1) 3c	Thermal impedance (see 4.3.2)
4	Not applicable
5	Not applicable
6	Not applicable
7a	Optional
7b	Optional
8	Not required
9	V_{F2} and I_{R1} (2)
10	Not applicable
11	Not applicable
12	See 4.3.3
13	Subgroup 2 of table I herein; V_{F2} and I_{R1}; ΔV_{F2} = .1 V (pk) from initial value; ΔI_{R1} = 100 percent of initial reading or ±5 µA dc, whichever is greater.
14a	Required
14b	Required
15	Not required
16	Not required

(1) Surge current shall precede thermal response. These tests shall be performed anytime after screen 3a and before screen 9.
(2) I_{R1} measurement shall not be indicative of an open condition.

4.3.1 Surge current. The test for surge current shall be in accordance with condition A of method 4066 of MIL-STD-750. The following details shall apply: I_O = 0; $V_{RM(W)}$ = 0; I_{FSM} = 500 A; six surges; T_A = 25 °C; tp = 8.3 ms; one minute minimum time between surges.

NOTE: A shorter time between surges will be allowed as long as it can be demonstrated that $\Delta T_J \leq 10$ °C, as measured immediately prior to the first current pulse and immediately after the final current pulse.

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4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3101 of MIL-STD-750. The thermal impedance conditions and maximum thermal impedance limit shall be derived by each vendor. The chosen thermal impedance measurement and conditions for each device in the qualification lot shall be submitted in the qualification report and a thermal impedance curve shall be plotted. The chosen thermal impedance values shall be considered final after the manufacturer has had the opportunity to test five consecutive lots. Heating current (I_H) ≥ rated I_O; t_H = 150 to 400 ms; t_{MD} = 50 to 300 µs; 50 mA ≤ I_M ≤ 250 mA.

4.3.3 Power burn-in. Power burn-in conditions shall be as follows: Method 1038 of MIL-STD-750, condition A, t = 48 hours.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-IVB (JAN, JANTX and JANTXV) of MIL-PRF-19500, and herein. Delta measurements shall be in accordance with table III herein.

Subgroup	Method	Condition
B2	1051	-55 °C to +175 °C, 25 cycles.
B2	4066	Condition A, T_C = 150 °C; V_R = rated V_R (AC peak) (see 1.3), I_f (surge) = 500 A; I_O = 35 A dc; six 1/120 second surges. 1 minute minimum time between surges. t_p = 8.3 ms. or Test condition C; T_A = T_C = T_J = 175 °C; V_R = rated V_R (AC peak) (see 1.3), I_O = 0; $I_{FSM(surge)}$ = 500 A; six surges 1/120 s, t_p = 8.3 ms, one minute minimum time between surges.
B3	1037	0.25 rated I_O ≤ I_O applied ≤ rated I_O (see 4.5.1) 2,000 cycles.
B3	1038	Condition A; 340 hrs. (Separate samples may be used.)
B5		Not required.

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4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table E-VII of MIL-PRF-19500, and herein. Delta measurements shall be in accordance with table III herein.

Subgroup	Method	Condition
C2	1056	0 °C to +100 °C, 10 cycles.
C2	2036	Test condition A, weight = 15 pounds (6.8 kg), t = 15 s.
C2	2036	Test condition D1, terminal torque = 3 pound-inches (0.339 N-m), t = 15 s.
C2	2036	Test condition D2, stud torque = 30 pound-inches (3.39 N-m), t = 15 s.
C5	3101 or 4081	$R_{\theta JC}$ = 0.8 °C/W maximum; I_M = 50 to 250 mA; $I_H \geq$ rated I_O; t_H = 25s; t_{MD} = 50 to 300 µs.
C6	1037	0.25 rated $I_O \leq I_O$ applied $\leq$ rated I_O (see 4.5.1) 6,000 cycles.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500, and table II herein. Delta measurements shall be in accordance with table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 DC intermittent operation life. A cycle shall consist of an on period, when power is applied suddenly, not gradually, to the device for the time necessary to achieve a delta case temperature of 85 °C +15 °C -5 °C followed by an off period, when the power is suddenly removed, for cooling. Auxiliary (forced) cooling is permitted during the off period only. 30 s $\leq t_{Heating} \leq$ 60 s, $P = V_F \times I_F$ or $P = V_{FPK} \times I_{av}$ if using sine wave current. DC full wave current (or equivalent half sine wave current) shall be used for the power required during the on period. The test power, or current, shall be at least the free air rating. Within the time interval of 50 cycles before to 500 cycles after the termination of the test, the sample units shall be removed from the specified test conditions and allowed to reach room ambient conditions. Specified end-point measurements for qualification and verification shall be completed within 96 hours after removal of sample units from the specified test conditions. Additional readings may be taken at the discretion of the manufacturer.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Forward voltage	4011	Condition A, t_p < 8.3 ms, duty cycle ≤ 2 percent pulse I_F = 110 A (pk)	V_{F2}		1.4	V dc
Reverse current	4016	Condition A or B, V_R = rated V_R (see 1.3)	I_{R1}		10	µA dc
Thermal impedance	3101	See 4.3.2	$Z_{\theta JX}$			°C/W
Subgroup 3						
High temperature operation:		T_C = 150 °C				
Reverse current	4016	Condition A or B, V_R = rated $V_{R(pk)}$ (see 1.3)	I_{R2}		1	mA dc
Subgroups 4 and 5						
Not applicable						
Subgroup 6 2/						
Forward voltage	4011	Condition A, $I_F = I_{f(surge)}$; t_p = 800 µs	V_{F1}		2.5	V dc
Forward voltage	4011	Condition A, $I_F = I_{f(surge)}$; t_p = 8.3 ms	V_{F1}		2.3	V dc
Subgroup 7						
Not applicable						

1/ For sampling plan, seen MIL-PRF-19500.
2/ V_{F1} shall be performed with either t_p = 800 µs or t_p = 8.3 ms.

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TABLE II. Group E inspection (all quality levels) for qualification and requalification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock	1056	100 cycles, 0 ° to +100 °C.	
Electrical measurements		See table I, subgroup 2 and table III herein.	
Subgroup 2			22 devices, c = 0
Steady-state dc blocking life	1038	1,000 hours, condition A; V_R = 80 percent of V_{RM}.	
Electrical measurements		See table I, subgroup 2 and table III herein.	
Subgroup 4			
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			22 devices, c = 0
Barometric pressure, reduced (altitude operation)	1001	Pressure (see 1.3); t = 1 minimum DC method; $V_R = V_{RM}$ (see 1.3); I_{R1} = should be monitored and shall not exceed table I limits.	
Subgroup 6			
Not applicable			
Subgroup 7			
Not applicable			

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TABLE III. Groups B, C, and E delta electrical end-point measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Forward voltage	4011	Condition A, I_F =110 a(pk) dc t_p < 8.3 ms, duty cycle ≤ 2 percent pulse.	ΔV_{F2}	±50 mV dc maximum change from previous measured value.		
2.	Reverse current	4016	Condition A or B, V_R = rated V_R (dc) (see 1.3)	ΔI_{R1} 4/	±100 percent of initial value or ±5.0 µA, (whichever is greater) from previous measured value.		

1/ The delta electrical measurements taken during group B inspection for quality levels JAN, JANTX and JANTXV (table E-VIB of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of step 1 shall also be taken.
   b. In addition to the measurements specified for subgroup 3, the measurements of steps 1 and 2 shall also be taken.

2/ The delta electrical measurements taken during group C inspection for all quality levels (table E-VII of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 2, the measurements of step 1 shall also be taken.
   b. In addition to the measurements specified for subgroup 6, the measurements of steps 1 and 2 shall also be taken.

3/ The delta electrical measurements taken during E inspection for all quality levels (table E-IX of MIL-PRF-19500) shall be as follows:
   a. In addition to the measurements specified for subgroup 1, the measurements of step 1 shall also be taken.
   b. In addition to the measurements specified for subgroup 2, the measurements of step 2 shall also be taken.

4/ Devices which exceed the table I limits for this test shall be rejected.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

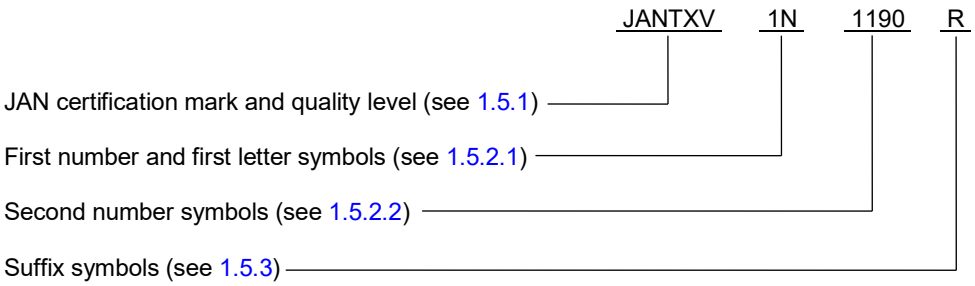
b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.4.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

6.4 PIN construction example. The PINs for encapsulated devices and are constructed using the following form.



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6.5 List of PINs. The following is a list of possible PINs available for encapsulated devices covered by this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level
Standard polarity		
JAN1N1184	JANTX1N1184	JANTXV1N1184
JAN1N1186	JANTX1N1186	JANTXV1N1186
JAN1N1188	JANTX1N1188	JANTXV1N1188
JAN1N1190	JANTX1N1190	JANTXV1N1190
JAN1N3766	JANTX1N3766	JANTXV1N3766
JAN1N3768	JANTX1N3768	JANTXV1N3768
Reverse polarity		
JAN1N1184R	JANTX1N1184R	JANTXV1N1184R
JAN1N1186R	JANTX1N1186R	JANTXV1N1186R
JAN1N1188R	JANTX1N1188R	JANTXV1N1188R
JAN1N1190R	JANTX1N1190R	JANTXV1N1190R
JAN1N3766R	JANTX1N3766R	JANTXV1N3766R
JAN1N3768R	JANTX1N3768R	JANTXV1N3768R

6.6 Amendment notations. The margins of this specification are marked with vertical bars to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Custodians:
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Review activities:
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